Beyond the Shadow of a Doubt: Benchmarking MCMC Sampling Algorithms for Multimodal Astrophysical Inference

Progress Report

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April 13, 2026

Abstract

As astrophysics enters the era of petabyte-scale datasets, the choice of inference algorithm and implementation for a research problem could make the difference between an intractable or incorrect analysis and a major scientific discovery. The high dimensionality and multi-modality of posterior distributions can pose problems for popular inference tools. To ensure that the astrophysics community is fully leveraging available inference algorithms, we benchmark a set of widespread tools from across the astrophysical literature. We benchmark eight inference algorithms—random walk Metropolis-Hastings (RWMH), affine-invariant ensemble sampling, parallel tempering, NUTS, SMC with tempering, non-reversible parallel tempering, ensemble Kalman sampling, and nested sampling (as a non-MCMC comparator)—on two test problems: a synthetic 8-component 2D Gaussian mixture with known ground truth, and a simulated exoplanet transit with stellar spots. All experiments run on AMD EPYC 9474F 48-core CPUs and Nvidia L40S GPUs using JAX to ensure implementation-agnostic comparison. We evaluate each sampler on mode recovery, posterior accuracy, and computational efficiency (ESS per unit compute time), and will produce a concrete set of recommendations for practitioners.

1 Project Overview

The central question of this project is: *Which parallelizable MCMC-family samplers most reliably recover multimodal posteriors in astrophysical settings, and at what computational cost?* We benchmark eight algorithms: RWMH, affine-invariant ensemble sampling [6, 4], reversible parallel tempering (SEO) [10], non-reversible parallel tempering (DEO) [9], NUTS, SMC with tempering [3], ensemble Kalman sampling [5], and nested sampling [8] as a non-MCMC reference.

We evaluate each sampler on two test problems. The first is an 8-component 2D Gaussian mixture with fully known ground-truth modes, weights, and covariances, enabling precise measurement of mode recovery and posterior accuracy. The second is a simulated exoplanet transit with stellar spots, a problem known to produce multimodal posteriors, which tests each sampler in a realistic astrophysical setting.

The primary evaluation criteria are: mode recovery (fraction of modes recovered, recovered mode weights), posterior accuracy (Wasserstein-2 distance from ground truth where available), and computational efficiency (ESS per log-density evaluation and ESS per wall-clock second). Secondary diagnostics include $\hat{R}$, acceptance rates, and—for parallel tempering variants—swap acceptance rates and round-trip counts between temperature levels.

The full implementation stack is JAX [1] / BlackJAX [2] / NumPyro [7], ensuring hardware-accelerated, implementation-agnostic comparison across all algorithms.

2 Division of Labor Update

The division of labor has been reshuffled relative to the original proposal. Lucas Arthur has taken ownership of both parallel tempering variants—reversible SEO and non-reversible DEO—in addition to his originally assigned algorithms (NUTS and SMC with tempering). This consolidation arose naturally because the two tempering variants share substantial infrastructure (temperature ladder management, swap kernels, diagnostic logging), and keeping them under a single owner reduced integration overhead.

Samson Mercier has taken on all forward model development, including implementation and validation of the stellar spot transit model against external transit-fitting codes, in addition to his originally assigned algorithms (RWMH, affine-invariant MCMC, and nested sampling).

The ensemble Kalman sampler (EKS) is deprioritized. Implementation is on hold until all other samplers have been run and results collated across both test problems. Samson Mercier will implement it if time permits; it will be omitted from the final comparison if it is not ready in time.

3 Progress to Date

3.1 Gaussian Mixture Model — Substantially Complete

Seven of the eight samplers have been fully implemented and run on the Gaussian mixture problem: RWMH, affine-invariant MCMC, reversible PT (SEO), non-reversible PT (DEO), NUTS, SMC with tempering, and nested sampling. EKS is the sole missing algorithm; it is deferred pending completion of transit model runs.

Each sampler produces the following outputs: posterior samples in a standardized format, corner plots, and a diagnostics record containing ESS, $\hat{R}$, acceptance rates, recovered mode weights, and inter-mode transition counts (for chain-based methods).

3.2 Stellar Spot Transit Model — Forward Model Complete, Sampling Not Yet Started

The transit forward model has been implemented using the `sajax` library with a NumPyro likelihood and a BlackJAX-compatible log-density interface. The inference problem is to recover three spot parameters—latitude, longitude, and radius—from a synthetic noisy light curve generated from a known ground-truth configuration. This is a reduced scope relative to the proposal, which described a 12-parameter model; the three-parameter version was chosen to keep the transit problem tractable while preserving the multimodal structure of interest. Samson Mercier has validated the forward model against independent transit-fitting codes.

Sampler scripts for the transit problem have not yet been written. Writing and running these scripts is the primary remaining task.

4 Preliminary Results: Gaussian Mixture

Detailed quantitative analysis is reserved for the final report; we summarize high-level qualitative observations here. Notable observations include:

- **RWMH** shows poor mixing between modes. The chain rarely crosses the low-density region separating mixture components, resulting in substantially fewer inter-mode transitions and lower ESS per evaluation than all other methods.
- **Non-reversible PT (DEO)** achieves substantially more inter-mode transitions than reversible PT (SEO) at matched compute budgets, consistent with the theoretical prediction that non-reversible temperature swaps improve round-trip rates [9].
- **Nested sampling** reliably recovers all eight modes. Its ESS per log-density evaluation is lower than gradient-based methods, but it provides an accurate evidence estimate and serves as a robust reference.
- **NUTS** and **SMC with tempering** show strong ESS per evaluation on this low-dimensional problem; their relative performance on the higher-dimensional transit model remains to be seen.

5 Revised Plan and Remaining Tasks

Table 1 lists the remaining tasks and internal deadlines.

Task	**Owner**	**Target**
Run all 7 implemented samplers on transit model	Both	Apr 25
Collate all results; build comparison tables	Both	Apr 25
Implement EKS (if time permits)	Samson	Apr 25
Produce final figures	Both	Apr 28
Write final report	Both	May 1

Table 1: Remaining tasks and internal deadlines.

6 Planned Figures and Evaluation Framework

Comparing the eight algorithms requires care because they produce fundamentally different outputs: correlated samples (RWMH, PT, NUTS), weighted particle sets (SMC, nested sampling), or continuously evolved ensembles (EKS, affine-invariant). We therefore report efficiency as ESS per log-likelihood evaluation, the most universally applicable unit of compute. For chain-based methods, warm-up evaluations are included in the denominator to give an honest accounting of total compute. The primary mode-recovery metrics are the per-mode weight error $|\hat{w}_k - w_k|$, total variation distance from the true mixture, and (for chain methods) inter-mode transition count. For the Gaussian mixture we also report Wasserstein-2 distance to ground truth; for the transit model we use nested-sampling output as a reference distribution.

Anticipated figures: (1) KDE posterior overlays for all samplers on the 2D mixture against the true density; (2) per-mode weight recovery bar charts; (3) ESS per likelihood evaluation across samplers and both problems; (4) convergence curves (Wasserstein distance vs. evaluations); (5) PT swap acceptance and round-trip rates for SEO vs. DEO [9]; (6) transit model corner plots per sampler; (7) a summary table of $\hat{R}$, MCSE, ESS/eval, and modes recovered across both problems.

References

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